MILITARY SPECIFICATION

ADHESIVE, NEOPRENE BASE, MEDIUM VISCOSITY

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers one type of medium viscosity neoprene base adhesive, using aromatic solvents as the vehicle.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, publications, and drawings required by suppliers in connection with a specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

FSC-1330

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PUBLICATIONS

American Society for Testing and Materials

ASTM D 1076	Concentrated, Ammonia Preserved, Creamed and Centrifuged Natural Rubber Latex
ASTM D 1084	Methods of Test for Consistency of Adhesives

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Preproduction samples. Unless otherwise specified (see 6.2), preproduction samples of the adhesive shall be manufactured using the materials and procedures proposed for production.

3.2 Data. No data are required by this specification, or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.3 Materials. This adhesive shall consist of a polychloroprene (neoprene) base and volatile aromatic solvents as the vehicle.

3.4 Physical and chemical properties

3.4.1 Total solids content. The solids content of the adhesive shall be no less than 18 percent and no more than 25 percent.

3.4.2 Viscosity. The viscosity of the adhesive shall be no less than 3,000 nor more than 4,000 centipoises.

3.4.3 Peel strength. The peel strength of the adhesive shall be no less than 17 pounds per inch of width.

3.5 Storage life. The storage life (see 6.3.1) of the adhesive shall be a minimum of 1 year from date of manufacture when stored at a maximum of 70° Fahrenheit (F) in the original, unopened container.

3.6 Workmanship. The material shall be uniform in quality and shall be free from impurities and other defects that could adversely affect its use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the adhesive shall be classified as follows:

a. Preproduction inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of sufficient quantity of adhesive manufactured in accordance with 3.1 and 3.3 shall be subjected to three each of the preproduction test detailed in 4.4 at an activity designated by the procuring activity. Further production of the adhesive by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Quality conformance inspection sampling. Except as otherwise specified, the sampling plans and procedures set forth in MIL-STD-105, inspection level S-4, shall be used in the determination of acceptability of the adhesive inspected as specified in 4.5.

4.3.3 Lot. A lot of adhesive shall consist of all the material from one batch or run, manufactured by one unchanged process by the same manufacturer, and submitted for inspection at one time.

4.4 Preproduction inspection. The preproduction sample, in addition to satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following test.

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4.4.1 Viscosity test. The viscosity shall be determined in accordance with ASTM D 1084, Method B.

4.5 Quality conformance inspection

4.5.1 Examination of workmanship. The contents of each container in a lot shall be visually examined to determine compliance with 3.6.

4.5.2 Storage life. The storage life shall be determined by an examination of the manufacturer's certification to the requirement of 3.5.

4.5.3 Packaging, packing, and marking. Visually examine each container in a lot to determine conformance with Section 5.

4.5.4 Total solids content test. The solids content shall be determined in accordance with ASTM D 1076.

4.5.5 Peel strength test. The peel test shall be determined by cutting cotton duck fabric into 12 strips four inches long by one inch wide. A coating of adhesive shall be applied by brush to three inches of each strip and allowed to dry for two hours. Then a second coat of adhesive shall be applied by brush and allowed to dry for two hours. Then a third coat of the adhesive shall be applied by brush and allowed to dry for 30 minutes. At this time the 12 strips shall be paired and laminated so the uncoated inch of each strip is matched against that of the other strip. The ensuing six laminations shall be pressed immediately on a laboratory press with 1,000 ± 100 pounds of pressure for 30 ± 5 seconds. The six laminations shall be allowed to age for three days or placed in an oven for 16 hours at 158 ± 5°F and cooled to room temperature before testing. Then using a Scott Tester Model U (or equivalent) with one-inch grips (which grip the uncoated section of the strips) and pulling at the rate of two inches per minute, the strips shall be pulled apart. The duck fabric used for this test shall be a number 10 canvas duck, which weighs approximately 14.7 ounces per square yard and has approximately 17 yarns per inch in warp and 26 per inch in fill.

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4.6 Acceptance criteria

4.6.1 Preproduction. Failure of any sample to conform to any requirements of this specification shall be cause for rejection of the preproduction lot.

4.6.2 Quality conformance. Acceptance shall be based upon an Acceptable Quality Level of 4.0 percent in accordance with MIL-STD-105, level S-4, normal inspection, single sampling.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified by the procuring activity, the preservation, packaging, and packing shall be in accordance with MIL-P-116, Method III.

5.2 Marking. Each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Number and revision letter of this specification
b. Manufacturer's and supplier's name and location
c. Material trade name and chemical name
d. Lot number and date of manufacture
e. Net weight or volume
f. Storage instructions if required
g. Safety instructions if required

6. NOTES

6.1 Intended use. The material purchased in accordance with this specification is intended to be used as an adhesive in rocket motors.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following information:

a. Title, number, and date of this specification
b. Type, grade, and class required (if applicable)
c. Whether preproduction samples are required (see 3.1)
d. Lot size (see 4.3.3)
e. Size of container
f. Place of delivery

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g. Responsibility for inspection (see 4.1)
h. Disposition of test data if source inspected
i. Minimum storage life required (see 3.5)
j. Data to be retained or submitted (see 3.2)

6.3 Definition

6.3.1 Storage life. Storage life is the period of time during which the packaged adhesive can be stored under specific conditions and meet the peel strength requirement of 3.4.3 of this specification.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-A-82569(OS) ADHESIVE, NEOPRENE BASE, MEDIUM VISCOSITY

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

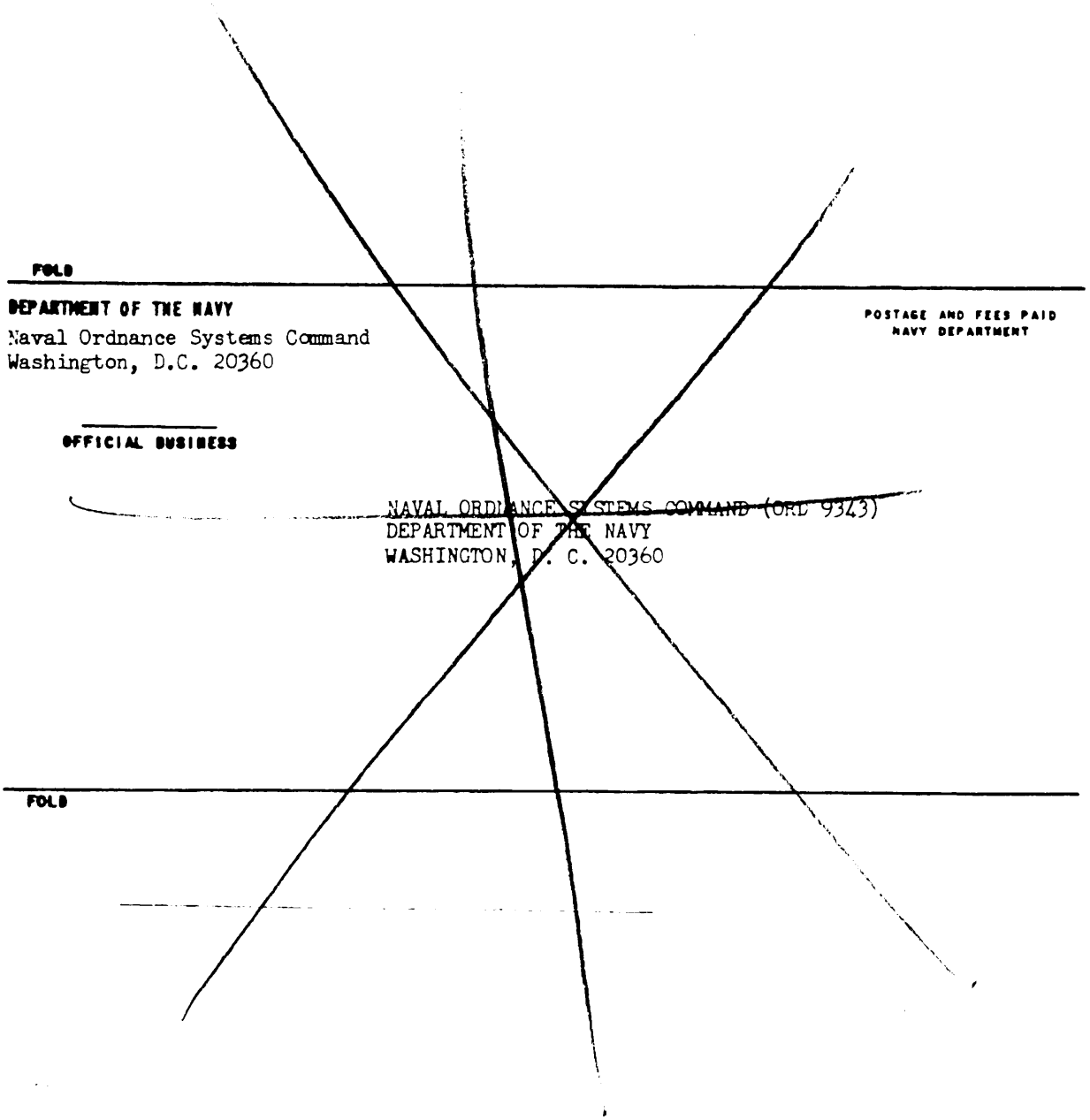
4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Ordnance Systems Command
Washington, D.C. 20360

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